CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 13: **Earth as a System: Energy, Matter and Life** | Quick revision, one page

Spheres of the Earth

- Geosphere: rocks, soil and interior. Hydrosphere: all water (oceans, rivers, ice, groundwater, vapour). Atmosphere: envelope of gases. Biosphere: all living organisms with their environment.
- Atmosphere layers from the ground: troposphere (weather), stratosphere (ozone layer), mesosphere, thermosphere. Composition of dry air: about 78% $\mathrm{N_2}$, 21% $\mathrm{O_2}$, 0.04% $\mathrm{CO_2}$.

Energy balance and albedo

- Sun is the main source of energy. Albedo $= \dfrac{\text{solar radiation reflected}}{\text{solar radiation received}}$ (fraction between 0 and 1).
- High albedo: snow, ice, clouds. Low albedo: oceans, forests, dark soil. Lower albedo $\rightarrow$ more heat absorbed $\rightarrow$ warmer surface.
- Poles are colder than the equator because sunlight strikes them at a slant and spreads over a larger area.

Greenhouse effect and ozone

- Greenhouse gases ($\mathrm{CO_2}$, $\mathrm{CH_4}$, $\mathrm{N_2O}$, water vapour) trap heat; natural greenhouse effect keeps the Earth warm; enhanced effect causes global warming.
- Effects of global warming: melting of ice and glaciers, sea-level rise, extreme weather, changes in rainfall and crops.
- Ozone layer absorbs harmful ultraviolet rays; depleted by CFCs (chlorofluorocarbons) from refrigerators, aerosols and so on.

Cycles of matter

- Water cycle: evaporation $\rightarrow$ condensation (clouds) $\rightarrow$ precipitation $\rightarrow$ run-off and infiltration. Carbon cycle: $6\,\mathrm{CO_2} + 6\,\mathrm{H_2O} \xrightarrow{\text{light}} \mathrm{C_6H_{12}O_6} + 6\,\mathrm{O_2}$ (photosynthesis); respiration, combustion and decay return $\mathrm{CO_2}$.
- Nitrogen cycle: nitrogen fixation by lightning and by bacteria (*Rhizobium*) in legume root nodules; nitrates absorbed by plants; denitrifying bacteria return $\mathrm{N_2}$.

Human impact and rainwater

- Rainwater collected = roof area $\times$ rainfall depth; $1\ \mathrm{m^3} = 1000$ L.
- Afforestation, reducing fossil-fuel use, recycling and avoiding plastic reduce human impact; burning crop residue causes air pollution and releases $\mathrm{CO_2}$.

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